

Institute of Actuaries of India

Subject ST8 – General Insurance: Pricing

October 2014 Examination

INDICATIVE SOLUTION

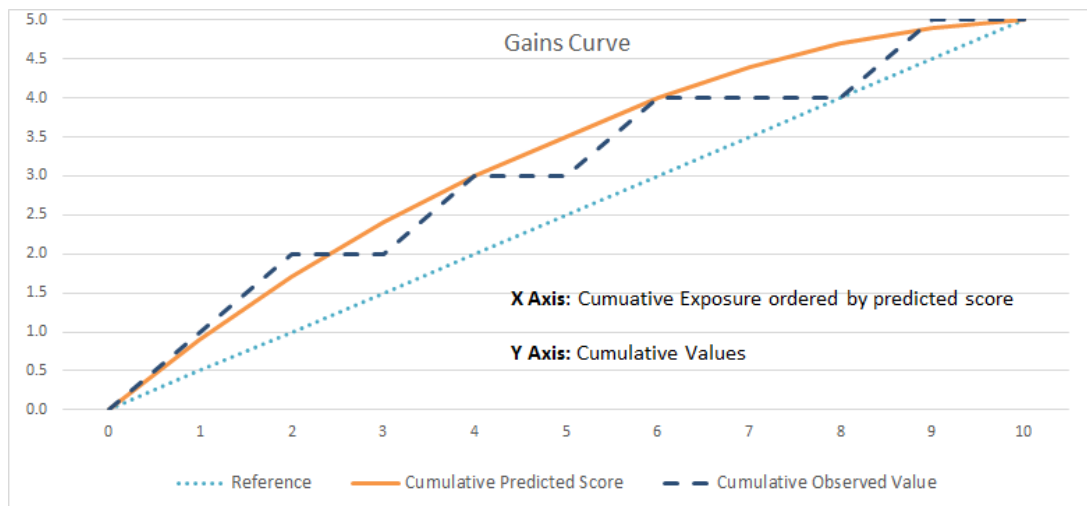
Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

In the data provided, data points are already sorted in descending order of predicted scores.

Data Point No.	1	2	3	4	5	6	7	8	9	10
Predicted Score	0.9	0.8	0.7	0.6	0.5	0.5	0.4	0.3	0.2	0.1
Observed Value	1.0	1.0	-	1.0	-	1.0	-	-	1.0	-
Cumulative Predicted Score	0.9	1.7	2.4	3.0	3.5	4.0	4.4	4.7	4.9	5.0
Cumulative Observed Value	1.0	2.0	2.0	3.0	3.0	4.0	4.0	4.0	5.0	5.0



[4 Marks]

Solution 2:

Various items:

- Proposed vs. Previous relativities – changes could impact retention rates, new business volumes and therefore, business mix
 - Use impact graphs to illustrate the impact on portfolio of moving to theoretically correct relativities
 - by indicating profitable and unprofitable business, impact graphs are useful in understanding the magnitude of existing cross-subsidies by considering the effect of all rating factors at the same
 - Unlikely that any dramatic changes, if suggested by the model, are implemented in practice
- Proposed vs. Competitor rates:
 - Consider restrictions, if any, imposed by any competition legislation
 - Model elasticity of customer demand to optimize prices subject to portfolio level constraints and stakeholder objectives
 - Identify if any particular cells are underpriced compared to other as it could lead to anti-selection – that is, comparison with competitors

[5 Marks]

Solution 3:

Let Y1 equals to total loss in 2012 and Y2 equal to total loss in 2013 and V1 equals to number of policies in 2012 and V2 number of policies in 2013.

Using the formulae for the mean and variance of the gamma distribution:

$$\mu(\theta) = E(X_k|\theta) = \frac{V_k}{V_k|\theta} = \theta$$

$$\sigma^2(\theta) = V_k[\text{Var}(X_k|\theta)] = V_k * \frac{V_k}{(V_k|\theta)^2} = \theta^2$$

Now using the formulae for the mean, variance and second moment of the uniform distribution:

$$\beta = E(\mu(\theta)) = E(\theta) = 22$$

$$\phi = E(\sigma^2(\theta)) = E(\theta^2) = 592$$

$$\lambda = \text{Var}(\mu(\theta)) = \text{Var}(\theta) = 108$$

We now use the Bühlmann-Straub formula for the credibility factor. The volume figure V is equal to the total exposure (i.e. total number of policies) over 2012 and 2013 which is 27 and the base statistic is the average claims ratio over the previous two years.

$$Z = \frac{V}{V + \frac{\phi}{\lambda}} = \frac{27}{27 + \frac{592}{108}} = 0.76096$$

$$= 0.83$$

$$X_3 = \frac{Y_1 + Y_2}{V_1 + V_2} = 22.59$$

So the Bühlmann-Straub credibility premium per unit of volume is:

$$C^{BE} = Z * X + (1 - Z) * \beta$$

$$= Z * 22.59 + (1 - Z) * 22$$

$$= 22.49$$

So the risk premium for the total losses in 2014 is = 247.41

[5 Marks]

Solution 4:

- i. The aggregate exceedance probability considers the probability that the aggregate losses from all loss events in a year exceeds a particular threshold. [1]
- ii. Inverse CDF of X_1+X_2 at, say, 'p' is not necessarily equal to CDFs of X_1 and X_2 at 'p', even when X_1 and X_2 are independent and identically distributed as well.

For example, if X and Y are i.i.d with $N[0,1]$, then:

Inverse CDF of both X and Y at p of 0.75 = 0.674

However, CDF of $X+Y$ which is $N[0,\text{SQRT}(2)]$ at p of 0.75 = 0.954; while $2*0.674 = 1.35$.

Inverse CDF of X_1+X_2 at, say, 'p' will be equal to CDFs of X_1 and X_2 at 'p' only when X_1 and X_2 have perfect positive correlation.

If they do not have perfect positive correlation, the inverse CDF would be lower, due to diversification benefits.

[3]

- iii. Take X_1^{th} simulation from Peril A and X_2^{th} from Peril B where X_1 and X_2 are i.i.d with uniform discrete distribution of integers from 1 to 10000. Now, add X_1^{th} simulation from Peril A and X_2^{th} simulation from Peril B. Run this, say, 100000 times, each time generating X_1 and X_2 from the same distribution independently. Sort them in descending order and select the 1000th value from that list.

OR

Random sort the observations from the outputs for Peril A and Peril B independently.

Add the amounts for Peril A and Peril B together, for each simulation. Sort these values in descending order and select the 1000th observation.

[2]

- iv. If the two perils are correlated positively, the loss threshold for the two perils could be greater than the amount estimated as above. This is especially true if the positive correlation is more prevalent in the tails of the distribution. [1]

[7 Marks]

Solution 5:

(i) The aim of run-off reinsurance is the transfer of reserve development risks. It provides cover against the insurer's earnings volatility arising from past activities. The entire book is sold to the reinsurer who assumes all remaining premiums and all of the risk. The claims reserves are also transferred from the insurer to the reinsurer.

It may be sought in circumstances such as:

- corporate restructuring
- mergers and acquisitions
- closing lines of business
- economic changes in the value of the liability
- regulatory, accounting or tax changes
- legal developments, for example court decisions.

[3]

(ii)

- The restoration of cover following a claim.
- There is often a limit to the number of times that cover may be reinstated, which means that there is a limit to the total recovery under the reinsurance.
- The terms of reinstatement will be specified in the contract.
- Once agreed, they are automatic and obligatory on both parties.
- To restore cover it may be necessary to pay a reinstatement premium
- This will usually be equal to the original premium of the reinsurance, multiplied by the proportion that the reinsurance recovery bears to the cover
- The amount of the premium may be pro-rated to the remaining term of the contract.

[4]

[7 Marks]

Solution 6:**Customers:**

Easy and quick comparison on premiums charged by various insurers; perhaps even the comparison of product features.

Would need some familiarity with product features to be able to come up with an informed decision.

Insurers:

Increased competition as most competitively priced quotes are likely to get converted.
Increased risk of anti-selection in cells that are under-priced and subsequently could lead to higher loss ratios

Greater volatility in UW results

Brings great visibility to small/new insurers with reduced spend on customer acquisition

Reduced customer lifetimes due to varying retention rates

Monitor data capture by aggregator and rating factors to be used
 Need to improve the internet sales platform of the company to deter customers from moving to the aggregators websites
 Need to constantly evolve product features and prices.

Intermediaries:

Competition to traditional agents
 Aggregators might find developing a sustainable revenue model a challenge
 Aggregators could monitor customer preference for various product features

Regulator:

Monitoring that product features and premiums are not misleading.
 Data confidentiality could be another area of concern

Shareholders / Investors:

Profitability and volatility concerns.

[8 Marks]

Solution 7:

- i. Probability of '6' turning up three times = $1 / (6 \times 6 \times 6)$

Let the required price per participant be P. Total Income = $10000 \times P$

Expected Outgo = Prize Money + Expenses = $\text{Binomial}(10000, 1/216) \times 100000 + 50 \times 10000$

$\Pr\{\text{Total Income} + \text{Surplus} > \text{Total Outgo}\} = 99\%$

$\Pr\{10000 \times P + 500000 > \text{Binomial}(10000, 1/216) \times 100000 + 50 \times 10000\} = 99\%$

$\Pr\{\text{Binomial}(10000, 1/216) < P/10 + 5 - 5\} = 99\%$

[Use Normal Approximation for $\text{Bin}[n, p]$ as $N[np, np(1-p)]$]

$\text{Binomial}(10000, 1/216) \sim \text{Normal}(10000/216, 10000 \times 215 / (216 \times 216))$

$\sim \text{SQRT}(10000 \times 215 / (216 \times 216)) \times N(0,1) + 10000/216$]

$\Pr\{\text{SQRT}(10000 \times 215 / (216 \times 216)) \times N(0,1) + 10000/216 < P/10\} = 99\%$

$\Pr\{6.79 \times N(0,1) + 46.3 < P/10\} = 99\%$

$\Pr\{6.79 \times N(0,1) < P/10 - 46.3\} = 99\%$

$\Pr\{N(0,1) < (P/10 - 46.3) / 6.79\} = 99\%$

$(P/10 - 46.3) / 6.79 = 2.326$

$P = 620.9$

[5]

ii.

Model uncertainty: When modelling, the risk that an inappropriate model has been used.

Parameter uncertainty: Risk from the model parameters that are inputs to the model whose exact values are unknown and therefore estimated

Process Uncertainty: risk inherent modelled figures will vary from the true value owing to random variation.

Model uncertainty: The model assumes that the roll of a dice is a random event with equal probability for all outcomes instead of modelling the physics behind the roll of the dice

Parameter Uncertainty: Number of participants is a parameter and the estimate of 10000 is not certain. As such, the probability of '6' turning up is also a parameter could be different from 1/6 if it is not unbiased.

Process uncertainty: Assuming that the model and parameters are actual, there is still a 1% chance of loss even if the expected profit is about 108 per participant.

[4]

[9 Marks]

Solution 8:

(a)

Inflation		
Adjustment		Inflated claim
1.34		347.08
1.28		185.06
1.22		309.95
1.16		300.98
1.10		253.58
1.05		298.20

adjustment for Rate	on level premium	Loss ratio
Change	rate	
105.93%	254.2401	136.52%
103.86%	166.17	111.37%
115.40%	300.0292	103.31%
106.85%	320.544	93.90%
101.76%	305.28	83.06%
106.00%	296.8	100.47%

The weighted average loss ratio is 103.15% and the average loss ratio is 104.77%.

[6]

(b) Various items to be considered:

- Rate changes could take place in the form of one-off events such as the implementation of a new rating structure.
- Even then, the rate changes may not be uniform across all segments.
- Rate changes could be different for new business and renewal business
- Especially, for commercial risks, it may not a one-off event as the rate change could vary by policy and not even at the segment level. Also, more subjectivity is involved and therefore more difficult to quantify rate change
- Changes to premium levels should be adjusted to exclude the impact of changes to risks, limits, deductibles, UW standards, exposure trend when it is sensitive to inflation etc.
- For commercial lines, the rate change is calculated at the policy level by comparing expiring premium to renewing premium after adjusting for items mentioned above
- Less straight forward in the case of new business or changes to business mix
- Different covers with different exposure bases, if any

[5]

[11 Marks]

Solution 9:

i.

- the (ground-up) loss frequency is independent of the limit purchased
- the (ground-up) severity is independent of the number of losses and of the limit purchased

[2]

ii.

Limit factors when there is an annual aggregate limit are more difficult to calculate than per-claim or per-occurrence factors. They require consideration of the frequency distribution of claims and/or occurrences. As noted above, under common assumptions, frequency is not considered in the calculation of simple per-claim or per-occurrence increased limits factors.

Often, ILFs for such cases are calculated in a two-step process. The first step is the calculation of per-occurrence or per-claim increased limits factors. As a second step, a simulation is run to evaluate the effects of multiple claimant or multiple occurrence situations.

[3]

- iii. In the Layer 250-500 with amount of 250 in the layer, the incremental ILF change is 0.13. While the incremental change is 0.27 for the layer 500-1000. Ratio of ILF change to layer amount is $0.13/250$ in layer 1 while it is $0.27/500$ which is more.

This is inconsistent with the property of ILFs that they increase with increasing limits, but at a decreasing rate.

[3]

iv.

PDF of X at $x = 90\% * (1/200)$ if $0 < x < 200$ 10% if $x = 200$

0 otherwise (1 mark)

$$\text{LEV at Limit 100} = \int_0^{100} x * f_x(x) dx + 100 * \Pr(X \geq 100)$$

$$= \int_0^{100} (90\%/200) x * dx + 100 * (1 - 90\%/2) = (90\%/400) * (100 * 100) + 55\% * 100 = 77.5$$

(1 mark)

$$\text{LEV at Limit 150} = \int_0^{150} x * f_x(x) dx + 150 * \Pr(X \geq 150)$$

$$= \int_0^{150} (90\%/200) x * dx + 150 * (1 - 90\% * 3/4)$$

$$= (90\%/400) * (150 * 150) + 32.5\% * 150 = 50.625 + 48.75 = 99.375 \quad (1 \text{ mark})$$

$$\text{LEV at Limit 200} = \int_0^{200} x * f_x(x) dx + 200 * \Pr(X \geq 200)$$

$$= \int_0^{200} (90\%/200) x * dx + 200 * (1 - 90\%)$$

$$= (90\%/400) * (200 * 200) + 10\% * 200 = 90 + 20 = 110 \quad (1 \text{ mark})$$

$$\text{ILF for Limit 150 with 100 as base limit} = \text{LEV}(150) / \text{LEV}(100) = 1.28$$

$$\text{ILF for Limit 200 with 100 as base limit} = \text{LEV}(200) / \text{LEV}(100) = 1.42 \quad (1 \text{ mark})$$

[4]**[12 Marks]****Solution 10:**

(i) Investigate following possible benefit of economies of scale:

- the benefit of economies of scale with the combined distribution network
- the savings that will be possible with combined model office
- the savings in underwriting, management of data and claims handling expenses
- the benefits/issues which may arise as an effect of combining the employee benefit schemes.
- benefits of possible diversification of business in terms of target market, location etc.

Investigate the trend in premium rates to identify any major change in the strategy. Check this in conjunction with following:

- market share and trend in the market share
- new business levels and trends

Investigate the following in relation to claims:

- frequency and average cost
- trends in above
- analyze the above by accident year, development year and payment year
- check for any large losses

Investigate the adequacy of reserves by approximately calculating the reserves, if sufficient information is available. Also investigate reserving trends in UPR and IBNR by comparing the reserves of various years to identify trend, if any.

Others:

- statutory solvency levels
- credit rating of the company
- free surplus
- the asset mix of assets supporting liability
- the asset mix of assets supporting statutory capital and free surplus

Investigate various accounting ratios/financial statements

- investment returns in conjunction with the asset mix
- profit margin
- ROCE
- P.E. ratio
- loss ratios by line of business
- operating expense ratio
- combined ratios
- any qualifications in the financial statements by the auditor
- any mention of contingent liabilities
- tax liabilities

[9]

(ii) Data

Policy related:

- Policy number
- Dates – inception/renewal/mid-term change
- Postcode
- Make/Model of the vehicle
- Safety features
- Age of vehicle/year of make
- Value
- Excess chosen (if there is an option)
- Age of policyholder
- Cover level e.g. act only or comprehensive
- Use of the vehicle
- Estimated annual mileage
- Add-ons chosen
- Premium amount

- Unique link between policy and claim, for matching
- Unique link between various policies e.g. unique customer identification id

Claims

- Incident date, reported date, settlement date
- Claim number
- Cause of loss
- Paid date for all claims payments
- Claim amount paid, by peril
- Claim amount outstanding, by peril
- Claims handling expenses
- Claim status

[6]**[15 Marks]****Solution 11:****(i) Other modifications to make it more effective:**

- This policy can be sourced for regular premium or single premium. However purpose of the policy might get defeated if the collection of regular premium in subsequent years drops. Therefore it is advisable to have only single premium mode of payment.
- To ensure that all the two wheelers buyers get insured for longer period, yearly renewable traditional policies shall not be allowed at the time of purchase of the vehicle.
- To further improve the same, regulator can ask insurers to offer only a long term product even if the policy is issued at a later stage.
- This will help improve the coverage period even for the current two wheelers' owners as they will be forced to purchase a long term plan.

[2]**(ii) Differences:**

- The policy term of traditional motor policy is of one year while policy term for this will be for a period of three or five years.
- Allowing for no claim benefit is not as straight forward as in the regular policy. The reason is that the entire premium might be collected at the start of the policy in the form of single premium.

Benefits to the insurer:

- As the premium is collected entirely at the policy issuance, this policy becomes more cost effective than a regular policy considering the premium size. Cost on premium reminder notice, renewal collection calling and other reminder systems can be saved. Cost of premium receipt and cover note/policy bond can also be saved.

- As the premiums are collected for entire five/three years; the overall premium collection will increase and therefore the overhead expense allocation per policy will reduce.
- Risk of lower renewal premium collection will not be there.
- All above benefits may allow insurers to price this cover at a lower premium and hence will further increase the take up rate at renewal stage after completion of five years.

Benefits to the policyholders:

- The hassle of paying regular premium will not be there.
- As mentioned above, if the overall premium levels get reduced, they will be paying lesser premium for the same benefit.
- As the entire premium is paid initially, policyholders might be able to fund this through the loan which they avail for purchasing the vehicle. This can help them pay the insurance premium as part of the EMI towards the loan.

Benefits to the distributor:

- As the premium levels in regular two wheelers cover are low, the resultant commissions are also low. In the proposed solution, entire premium will be collected upfront and therefore the commission level will also go up.
- Also there is a possibility that the insurers may pass on some savings to the distributors by way of higher commission rate.

[7]

(iii)

- **Pricing:**
 - i. As the cover is for three/five years, there is a need to estimate the expected pure risk premium for entire period of three/five years
 - ii. The expense loading need to revisited considering the fact that the policy is for three/five years
 - iii. There is also a need to allow for investment income while calculating the premium to be charged
- **No claim benefit:**
 - i. As the premium for all five years is collected at the start, no claim benefit can be given in the form of increasing the sum insured or extending the period of cover for policies which don't claim.
 - ii. So for no claim policies, the reduction in "insured declared value" can be lower as compared to policies with claim.
 - iii. Or for each year of policy year with no claim, cover of extra 3/6 months can be extended at the end of the policy term.
 - iv. Also, at the end of the policy term, for policies with lower than specified number of claims charge lesser premium for new policy.
 - v. For each year of policy year with no claim, reduce the deductible level.

- vi. For each year of policy year with no claim, provide discounted servicing or PUC Certificates.
 - vii. The NCB has to be in line with the competition as this will impact portability.
- Reserving
 - i. No change is expected in estimating IBNR and companies can continue following the current methods of allowing/estimating IBNR
 - ii. As it is long term policy for three/five years, UPR will need to be estimated for the outstanding period of cover for each policy on the date of valuation.

[6]

(iv)

- Increase deductible with increase in number of claims e.g. in case of first claim in a policy year deductible of 1,000, for second claim deductible of 2,000 and so on.
- Cap the no of claims per year and/or in the entire policy term e.g. maximum 2 claims in any policy year and/or 5 claims in 5 years.
- Limit the repairs done only in garages with cashless tie-up.
- Set up tight internal controls to ensure good premium rating, underwriting and claims control

[2]

[17 Marks]
